Ghalih Ahmad Salamun 陳家豪

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SUMMARY

Robotics & AI Engineering graduate with hands-on experience designing and building AUVs/ROVs for national and international competitions. Skilled in CAD, CFD, and mechanical design, with leadership roles in campus robotics and engineering communities.

Areas of Interest: Robotics, Autonomous Vehicles, Industrial Automation

EDUCATION

Universitas Airlangga | Robotics and Artificial Intelligence Engineering ***Aug 2022 - July 2026***
Graduated with the Faculty's Most Outstanding Graduate distinction ***GPA: 3.21/4.00***

EXPERIENCE

MARE VICTOR - Head Mechanic - Surabaya ***Nov 2023 - Nov 2024***

MARE VICTOR is a team I co-founded to research and produce a competition-ready Autonomous Underwater Vehicle (AUV) and Remote Operated Underwater Vehicle (ROV).

- Led a small team of 2 mechanics, providing instruction in basic CAD and CAM.
- Coordinated design and assembly efforts with the Electrical and Programming divisions to ensure full system integration.
- Manufactured and built a functioning ROV prototype within a constrained timeline of 11 days, using it to gain practical experience in underwater vehicle design, fabrication, and waterproofing.
- Built upon the experience and design knowledge from the ROV prototype to develop a competition-ready AUV for the Indonesia Underwater Robotics Contest in 2024.
- Drafted components using Autodesk Inventor and simulated the capabilities of the AUV up to 5 m in depth using Autodesk CFD.
- Conducted tests regarding the waterproofing and safety of the AUV to ensure total safety throughout the competitions.
- Fully assembled the AUV with dimensions of 55 x 44 x 24 cm.

MARE VICTOR - Senior Mechanic - Surabaya ***Nov 2024 - July 2026***

- Mentored a team of junior mechanics in AUV mechanical design, manufacturing, assembly, and testing.
- Reviewed and evaluated mechanical designs and manufacturing processes with the Head Mechanic to improve AUV performance as part of the preparation for Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2025.
- Collaborated with Electrical and Programming teams to integrate and optimize the AUV for SAUVC 2025.
- Drafted parts for making the AUVs using Autodesk Fusion, and analyzed and simulated the capabilities of the AUV using Autodesk CFD.
- Oversaw tests regarding the waterproofing and safety of the AUV.
- Manufactured and assembled a completely functioning AUV prototype within 2 months under extreme resource constraints (no allocated funding), securing 21st place out of 45 at SAUVC 2025.

Integrated Mechano-biosystem Lab, National Cheng Kung University - Research Intern ***Aug 2025 - Dec 2025***

- Developed a semi-autonomous robotic microtissue handling system integrating a DOBOT MG400 robotic arm, computer vision, and a custom suction-based end-effector.
- Designed and fabricated a 1 cc syringe-based pipette end-effector for controlled pickup and transfer of microtissues between Petri dishes and well plates.
- Implemented YOLOv8-based object detection and homography transformation to identify microtissue positions and convert image coordinates into the robot's workspace.
- Performed system calibration and experimental validation, achieving 0.853 precision, 0.921 recall, 0.886 F1-score, and 0.922 mAP@0.5 for microtissue detection.
- Evaluated robotic positioning accuracy through repeated experiments, achieving a 1.424 mm mean error and 1.881 mm RMS error.
- Continued the research as an undergraduate thesis focused on automated microtissue handling for laboratory automation and cancer drug-testing applications.

ORGANIZATION

Komunitas Robotika UNAIR - Treasurer ***Mar 2024 - Feb 2025***

- Document the earnings and spending of KOMBO UNAIR.
- Manage and prepare proposals for funding up to 135.000.000,00 IDR for research and competitions.

ACHIEVEMENTS

9th Place - Indonesia Underwater Robotics Contest (KRBAI) 2024

Held by the Indonesian Ministry of Education and Culture.

- Designed and built an Autonomous Underwater Vehicle (AUV) capable of navigating competition obstacles and gates, completing 3 of 4 tasks in 1 minute 22 seconds.
- Achieved 9th place nationally among 18 participating universities as part of MARE VICTOR.

Bronze Medalist - 37th National Student Scientific Week (PIMNAS) 2024

Held by the Indonesian Ministry of Education and Culture.

- Engineered a portable digital microscope prototype for monitoring CD4 levels in HIV patients using a microfluidic optical biosensor and IoT-based monitoring system.
- Presented the research and prototype before a national panel of judges, earning a Bronze Medal in the PKM-Karsa Cipta category.

21st Place - Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2025

Held by the Oceanic Engineering Society (OES) IEEE Singapore.

- Designed and built an AUV capable of autonomous navigation, object retrieval, and payload deployment, including a custom scissors-based retrieval mechanism.
- Achieved 21st place among 45 international teams while developing the vehicle with zero allocated funding.

Faculty's Most Outstanding Graduate - Universitas Airlangga, August 2026

Given by Universitas Airlangga.

- Chosen as the Most Outstanding Graduate of the Faculty of Advanced and Multidisciplinary Technology out of 224 graduates.

SKILLS

Computer-Aided Design (CAD) : Autodesk Fusion 360, Autodesk Inventor
Computer-Aided Manufacture (CAM) : Laser-cutting, 3D Printing
Computational Fluid Dynamics (CFD) : Autodesk CFD
Programming Language : Python
Computer Vision : YOLOv8, Homography Transformation

LANGUAGE

Indonesia : Native
English : Fluent
Test of English for International Communication (TOEIC) 18 Aug 2026.
Listening: 495/495 | Reading: 475/495 | Total Score: 970/990